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A Retrieval-Augmented Chatbot for the EDAB6808 Learner Guide

Executive Report

EDAB6808 Project Group — October 2026

Executive Summary

The project developed a guide-specific chatbot in Google Colab using PyMuPDF, BGE embeddings, Qdrant Local, hybrid retrieval, a Qwen2.5-3B-Instruct generator adapted with LoRA, and a Gradio interface. Its purpose is to answer students' questions using the EDAB6808 Learner Guide, identify the supporting source, and refuse requests that the guide cannot support. The assessed knowledge source remains the learner guide alone.

The guide was converted into 35 chunks, of which 33 were eligible for retrieval. Each chunk retains its text and source metadata. Semantic search and BM25 keyword search supply candidates, Reciprocal Rank Fusion combines their rankings, and a cross-encoder selects evidence. The final configuration passes up to four chunks to the generator and includes additional handling for questions that require more than one source.

Principal results from the documented Colab T4 run

Required source-page coverage at $k = 4$	25/25 answerable questions
Automatic answer-check pass rate	92%
Citation-page-check pass rate	76%
Unsupported questions refused	6/6
Median response time across the test set	3.4 seconds

The results separate evidence access from evidence use. Retrieval located the expected pages in every answerable test, yet the generator did not consistently combine or cite all required passages. LoRA increased the overall automatic answer-check pass rate from 88% to 92%, but multi-section performance fell from 80% to 60%, and citation-page matching fell from 80% to 76%. The adapter therefore improved some behaviours without producing a uniform quality gain.

Recommendation. Retain the present system as an assessed prototype, prioritise training examples that require multiple sources, and introduce claim-level answer and citation review before wider use. The 31-question test provides an initial assessment, not a general reliability guarantee.

1 Purpose, Scope and Assignment Alignment

1.1 The information problem

Students need to find assessment requirements, lecturer details and module arrangements across a structured PDF. A conversational interface can reduce the effort of locating this information, provided it keeps answers tied to the document. The assignment therefore requires accurate, transparent responses and explicit refusal when the guide does not supply the requested fact [**?**].

Retrieval-augmented generation supplies selected document passages to a language model before it answers. This connects generation to an external evidence store [Lewis et al., 2020]. In this project, the approach supports a narrow institutional task: explaining the supplied learner guide. External papers and documentation inform the implementation and this report; they are not added to the chatbot's assessed knowledge base.

1.2 Implemented components

Assignment requirement	**Project implementation**
Colab and source ingestion	Google Colab with a T4 GPU; PyMuPDF extraction of the learner-guide PDF.
Chunking and metadata	Page- and heading-aware chunks, stable IDs, original text and printed/PDF page labels.
Open-weight embeddings	BAAI/bge-small-en-v1.5, producing 384-dimensional vectors.
Vector database	Qdrant Local with stored text, source metadata, deterministic IDs and a Drive backup.
Relevance improvement	BM25, rank fusion, cross-encoder reranking, sub-question retrieval and page diversity.
Fine-tuned local generator	Qwen2.5-3B-Instruct with a LoRA adapter and evidence-only prompting.
Citations and abstention	Passage-ID mapping in Python, a relevance gate, model refusal and numerical flags.
Evaluation and interface	31 test questions, 12 development questions, comparison experiments and Gradio.

Table 1: Alignment of the implemented system with the required stack and tasks.

1.3 Success criteria and intended use

The project aims to retrieve the relevant passages, answer the requested facts, provide an appropriate citation and refuse unsupported questions. These are distinct outcomes: a correct page can contain an irrelevant passage, a valid passage ID can accompany an incomplete answer, and a well-written response can still be unsupported. Evaluation must therefore examine each stage separately.

The prototype is suitable for guide-based question answering and classroom demonstration. It does not establish official university policy beyond what the supplied guide states. An absence of information is reported as such rather than filled with a general assumption. Neither institutional use at scale nor reliable handling of all possible student wording has been demonstrated.

2 Source Preparation and Chunk Design

2.1 Extraction and table reconstruction

The 16-page learner guide is the only assessed knowledge source [Combrink, 2025]. The documented extraction produced 27,278 characters. PyMuPDF retains line text, font information and position, allowing the notebook to identify headings and recurring margin artefacts. Four tables were reconstructed: lecturer information, learning units, the module schedule and assessment weights.

Table handling protects the relationship between a label and its value. For example, the row for Assignment 5 must retain its 20% contribution beside the assessment label, rather than separating the two during PDF reading. Rows are represented with their column headings so they remain understandable when retrieved independently.

Cleaning removed 1,024 characters, approximately 3.8% of the extraction, including 15 page-number lines and 32 stray bullet symbols. The notebook restricts header/footer removal to repeated margin content or page numbers, protects table rows, and checks coverage before indexing. These safeguards reduce accidental content loss; they do not replace inspection of the extracted text.

2.2 Chunking trade-off

The selected configuration uses a maximum of 220 words and overlap of up to 40 words. Chunks remain within their page and heading segment. Sentence and table-row units are used where feasible; exceptionally long units have a word-window fallback. The documented result is 35 chunks, six containing table rows, with a median of 102 words. The table of contents and a heading-only stub are indexed but excluded from retrieval, leaving 33 searchable chunks.

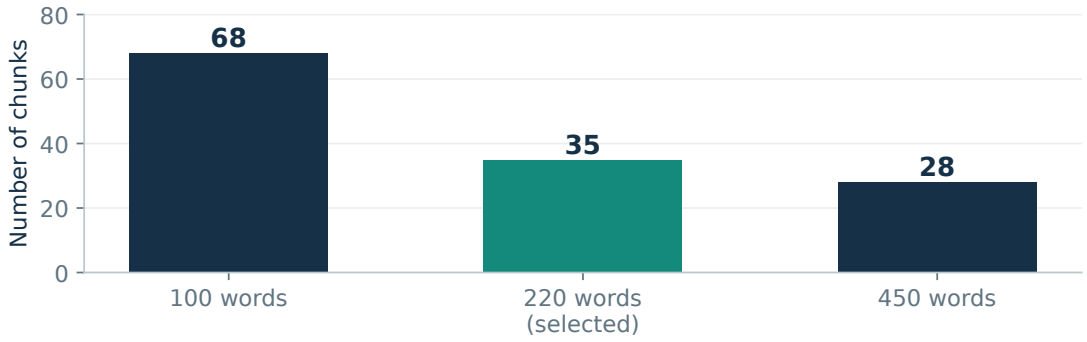


Figure 1: Chunk counts under three reported size limits. Larger limits reduce fragmentation, but the decline is limited because many sections are already short.

Each chunk carries a stable ID, text, PDF page, printed page, section/subsection and content type. Unit names are attached to schedule rows that otherwise refer only to a unit number. This helps a query about “Bayesian Networks” retrieve the corresponding schedule entry. Coverage assertions check that each source unit is represented and that the cleaned page text is retained through segmentation. Page numbering explicitly accounts for the unnumbered cover.

3 Architecture and Evidence Retrieval

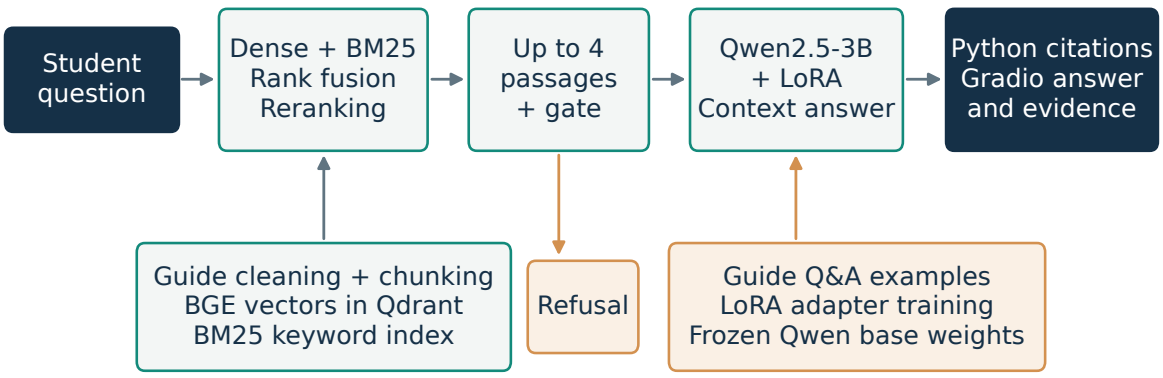


Figure 2: Implemented preparation, retrieval and generation flow. LoRA training is separate from BGE embedding; both contribute to the final question-answering pipeline.

3.1 Embedding and persistent storage

BGE encodes the heading path, passage text and applicable unit references into a normalised 384-dimensional vector [BAAI, n.d.]. The question uses the same encoder with a retrieval prefix. Qdrant compares dense vectors using cosine similarity and stores the original text and metadata in each point's payload [Qdrant, n.d.]. The 35 indexed vectors correspond to the 35 chunks, including the two retrieval-excluded records.

Deterministic point IDs prevent duplicate insertion on rerun. An index fingerprint covers the PDF, embedding model, chunk configuration and chunk text. A matching index is reopened; a changed configuration triggers rebuilding. The working database resides on the Colab disk, with a backup on Google Drive for restoration after a reset.

3.2 Hybrid selection and context coverage

Dense search returns up to 15 semantic candidates, while BM25 returns up to 15 keyword candidates over the searchable chunks. Reciprocal Rank Fusion combines rank positions with the notebook's constant of 60, avoiding the need to add incompatible raw scores [Cormack et al., 2009]. A cross-encoder, `ms-marco-MiniLM-L-6-v2`, scores question–passage pairs before final selection [Sentence Transformers, n.d.].

For multi-part questions, the system also searches sub-questions and reserves coverage for their strongest evidence. The default final context is four passages, with at most two from one page. This makes room for sources from different parts of the guide. Similarity and reranker scores are selection signals, not probabilities that an answer is correct. The architecture exposes weaker retrieval modes for comparison rather than assuming every added step must improve every metric.

4 Qwen Adaptation, Grounding and Citations

4.1 Generation and LoRA training

Qwen2.5-3B-Instruct is loaded locally through Transformers [Qwen, n.d.]. The prompt supplies only selected guide passages and requires concise answers, passage IDs and accurate copying of names, codes and numbers. It instructs the model to identify uncovered parts of a question and to return a fixed refusal when none of the requested information is supported.

LoRA keeps the pretrained base weights frozen while training low-rank weight updates [Hu et al., 2021]. The notebook targets the attention projections with rank 16, alpha 32 and dropout 0.05. It uses two epochs, a learning rate of 2×10^{-4} and a maximum sequence length of 2,048 tokens. Prompt tokens are masked in the training labels, so the learning objective focuses on the target response.

Training item	**Recorded value**
Synthetic question–answer pairs generated	66
Pairs rejected by the answer-word-overlap filter	23
Pairs rejected as too similar to test/development questions	2
Grounded pairs retained; refusal examples added	41; 10
Training examples; validation examples	46; 5
Trainable parameters; share of total parameters	approximately 7.4 million; 0.24%
Training time; final validation loss	3.7 minutes; approximately 0.25

Table 2: Reported adaptation data and training results. The filtering test uses lexical overlap, not a proof of factual support. *Source: recorded run summarised in* **?**.

Each supported training example contains one source passage and three distractors. Refusal targets use unrelated context. Similarity filtering reduces direct question overlap with the evaluation set but cannot establish complete semantic independence. The adapter, training records and logs are saved for reuse. Lower validation loss describes the training objective; it is not sufficient evidence of better end-to-end answers.

4.2 Citation mapping and refusal

Python maps recognised passage IDs to stored page and heading metadata. It filters IDs against the supplied evidence and discloses a top-ranked-passage fallback when no valid IDs are returned. A fallback provides a candidate source, not verified support. The relevance gate can refuse before generation; the model can also refuse after reading the context. Numerical checks flag answer values absent from the evidence/question text, but do not verify units, relationships or meaning. These controls support review without guaranteeing faithfulness.

5 Evaluation Design and Measurement

5.1 Question coverage and separation

The evaluation contains 31 test questions: 20 direct factual questions, five multi-section questions and six unsupported questions, spanning eight guide sections. This exceeds the brief's minimum of 12 questions across five sections. Four unsupported questions are close to the guide's subject matter, such as a pass mark or late-submission penalty that the guide does not state. Each test records expected pages and reference excerpts. A separate 12-question development set calibrates the relevance threshold.

Measure	Meaning in this notebook
Retrieval Hit@k	Expected-page matching within the retrieved list. Direct questions require any expected page; multi-section questions require all expected pages.
Mean reciprocal rank	Mean reciprocal rank of the first retrieved expected page. It does not measure whether all multi-part evidence is supported.
Answer accuracy	Automatic expected-keyword matching on answerable questions, with optional manual overrides. This is reported here as an answer-check pass rate.
Citation accuracy	Expected-page matching in the cited sources, rather than claim-level verification of each answer sentence.
Abstention success	Proportion of unsupported questions refused. False abstention measures refusals on answerable questions.
Response time	End-to-end answer-function time, including retrieval and, when invoked, generation; summarised across all 31 tests.
Context checks	Fuzzy reference-passage coverage and a rank-weighted average-precision-style score computed from retrieved text.

Table 3: Metric definitions checked against the evaluation code. The labels should be interpreted within these operational definitions.

5.2 Calibration and complementary controls

The documented selected threshold is -4.0, with development balanced accuracy of 0.83. This accuracy averages the retention of answerable questions and the blocking of unsupported questions. Scores overlap: an answerable development question scored as low as -6.09, while an unsupported one scored as high as -0.58. A single cutoff therefore cannot perfectly separate the two groups. Model refusal remains a complementary control after the gate.

RAG assessment should distinguish retrieval, answer quality and faithfulness [Es et al., 2023]. The walkthrough reports that the RAGAS package failed to import, and the notebook provides RapidFuzz alternatives. Its reported context recall of 1.00 and context precision of 0.93 are *RAGAS-style fuzzy metrics*, not successfully executed RAGAS package scores. They should not be treated as human relevance judgments or standard unweighted precision.

6 Retrieval Experiments and Context Trade-offs

6.1 What the comparison shows

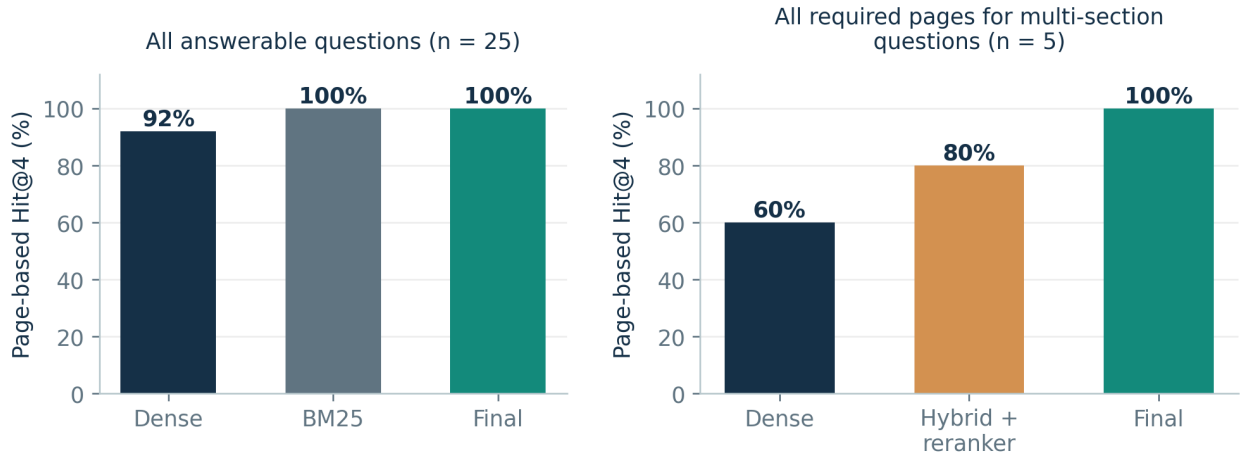


Figure 3: Reported source-page coverage for selected retrieval modes. The multi-section panel requires all expected pages. Only values explicitly stated in the walkthrough are plotted.

Dense retrieval reached 92% overall page coverage and 60% all-page coverage on multi-section questions. Hybrid retrieval with reranking raised multi-section coverage to 80%; the final coverage controls reached 100% on both measures. The final mean reciprocal rank was 0.96.

BM25 alone also reached 100% overall Hit@4. Guide-like wording may favour keyword retrieval, so a general hybrid advantage has not been established. Additional paraphrase tests are needed. Hit@1 decreased from 96% under BM25 to 92% under final selection when sub-question coverage was prioritised.

6.2 Why four passages and 220 words were retained

At one passage, the k sweep reached 72% overall coverage and missed all multi-section questions. Coverage reached 100% from two passages onward. Four were retained as a context margin, with approximately 790 evidence tokens. This does not establish a universal optimum or a generation benefit from additional context.

Chunk limit	Page-based Hit@4	Approximate evidence tokens
120 words	92%	Not specified in the summary
220 words	96%	762
350 words	100%	1,090

Table 4: Separate chunk-size experiment using hybrid retrieval and reranking without sub-question expansion or the page cap.

The 350-word setting increased estimated context length by approximately 43%. The project retained 220 words because the final coverage controls already achieved full page coverage. These token estimates use word counts, not exact Qwen tokenisation or memory measurements.

7 Answer Quality and the Effect of LoRA

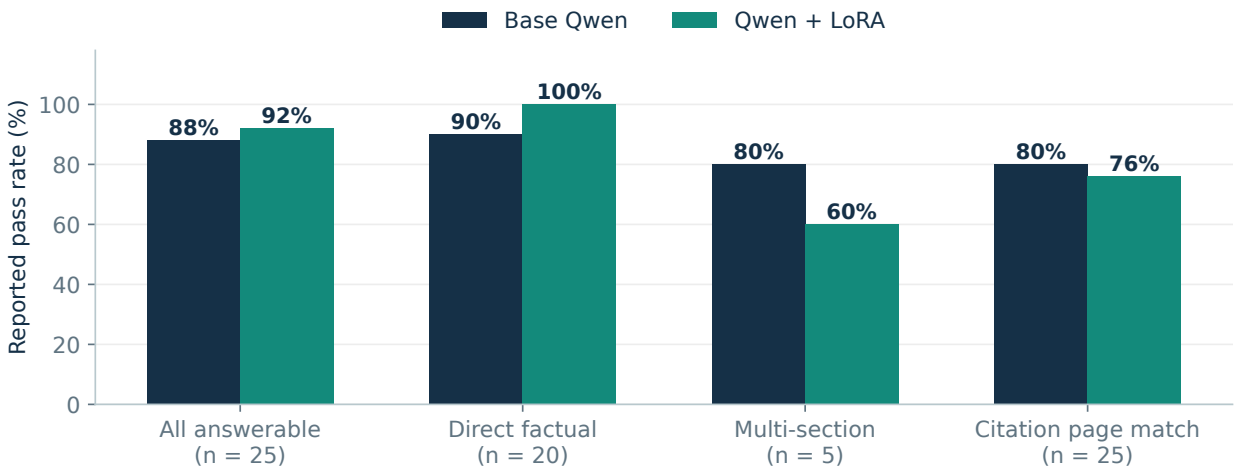


Figure 4: Base and adapted Qwen on the same test questions and retrieval configuration. Answer measures use keyword checks; citation measures use expected-page matching.

The reported overall answer-check pass rate improved by four percentage points, from 88% to 92%. With 25 answerable questions, this corresponds to 22 versus 23 automatic passes. Direct factual passes increased from 18/20 to 20/20. Multi-section passes decreased from 4/5 to 3/5. The net improvement is therefore small and masks a deterioration in the task of combining sources.

Citation-page matching declined from 20/25 to 19/25. Every adapted answer supplied a recognised passage ID, compared with five base answers that required fallback citations. Improved citation formatting did not imply improved source selection. The walkthrough identifies a response about weekly class hours that gave the correct fact but cited PDF page 12 rather than the expected page 6. It also identifies multi-part answers that cited the schedule while omitting a required assessment source.

Additional result	**Base**	**LoRA**
Answers needing fallback citations	5	0
Answered questions with a flagged unsupported number	8%	0%
Median response time, all test questions	2.6 s	3.4 s

Table 5: Additional recorded comparison outcomes. A numerical flag is a limited string check, not complete factual verification.

All six unsupported questions were refused in the adapted pipeline, five by the relevance gate and one by the model. No answerable test was refused. The 90th percentile response time was 5.2 seconds. Because timing includes unsupported requests that may bypass generation, it should not be interpreted as the generator's latency alone.

The observed pattern is compatible with the single-source design of the training targets: it may favour concise factual responses and single-passage citations. This is a hypothesis rather than an experimentally isolated cause. Multiple runs and controlled training-data comparisons are needed before attributing the changes to a specific training mechanism.

8 Limitations, Recommendations and Conclusion

8.1 Limits of the evidence

The evaluation has only five multi-section and six unsupported questions. One multi-section outcome changes that subgroup's rate by 20 percentage points. Guide-like wording favours lexical retrieval. Keyword checks can pass incomplete or contradictory answers, and page checks can accept an irrelevant passage from the expected page.

The small synthetic training set uses one positive source plus distractors. Lexical overlap does not prove entailment, and question-similarity filtering cannot eliminate all semantic overlap with evaluation. Numerical screening also considers the question and cannot verify a value's meaning, unit or role.

The numerical evaluation is documented in the walkthrough, while the archived notebook has no saved outputs. Raw exports and training logs were unavailable for reconciliation. The percentages are documented project results with the stated definitions; they have not been independently reproduced.

8.2 Prioritised improvements

Priority	Action	Purpose
1	Review each factual claim against its cited passage; add numeric boundary and unit checks.	Prevent keyword/page matches from overstating correctness.
2	Add two-source training targets and require citations for each requested part.	Improve complete answers and source coverage on multi-part questions.
3	Expand the test set with paraphrases, missing facts and misleading near-domain requests.	Assess robustness beyond wording close to the guide.
4	Preserve raw outputs, adapter metadata and compatible package versions; repeat timed runs.	Strengthen reproducibility and operational confidence.

Table 6: Recommended next steps ranked by their relevance to the observed weaknesses.

8.3 Demonstration and operational use

Gradio shows answers, citations and an evidence panel with page, heading, chunk ID, score and response time. Its share link forwards requests to the running application [Gradio, n.d.]; Colab must remain active. Before demonstration, restore the index, load the adapter and test the link. Keep recorded outputs as a fallback. A saved working configuration can be demonstrated without repeating training and evaluation.

8.4 Conclusion

The project implements the required RAG stack and a transparent guide-specific interface. Its strongest documented results are source-page retrieval and refusal of unsupported requests. Accurate use and citation of multiple sources remain the main challenge. Further validation is necessary before wider student-advising use.

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